Eric Walker

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EDUCATION

University of Colorado Boulder **August 2023 - May 2027**

Bachelor of Science in Electrical and Computer Engineering | Minor in Computer Science

- **GPA:** 3.61
- **Relevant Courses:** Circuits and Electronics; Circuits as Systems; Microelectronics; Digital Logic; Embedded Systems; Embedded Software; Database Systems; Computer Systems; Data Structures; Operating Systems; Computer Organization; PCB Design; Embedded AI

TECHNICAL WORK EXPERIENCE

Colorado Space Grant, CARMA Tech Team: **May 2026 - Present**

- Built, calibrated, and deployed NASA STELLA open-source spectrometers to capture visible and near-infrared (NIR) irradiance telemetry for research into plant heat-stress detection.
- Designed and built an enclosure for an environmental monitoring pod; assembled, programmed, and tested custom ESP32-S3 based sensor PCBs for soil off-gassing measurements.

TECHNICAL PROJECTS

Neural Amp Modeling Guitar Amplifier: **August 2026 - Present**

- Developing a portable, touchscreen-controlled guitar amplifier that uses Neural Amp Modeler to run multiple amplifier profiles and effects within a single embedded system.
- Led development and validation of a touchscreen interface using Embedded C and LVGL on an ESP32-S3, supporting amplifier selection, adjustable effects, presets, and touch-based signal-chain reordering.
- Contributed to system architecture, component selection, and custom PCB development for a dual-processor design separating the ESP32-S3 interface from Daisy Seed3 audio processing.
- *Skills Used: Embedded C, ESP32-S3, ESP-IDF, LVGL, System Architecture, Component Selection, Hardware Design*

Wi-Fi-Controlled Discrete Circuit Robot: **March 2026**

- Designed and built a closed-loop robotic control system powered by an Arduino Nano Every, integrating dual speed sensors and hardware compensator circuits for precise, programmed movement control.
- Constructed discrete motor driver circuitry using MOSFETs and BJTs for bidirectional movement, utilizing encoder pulse tracking to execute exact rotational and linear movements.
- Integrated an ESP-01 module utilizing ESP-NOW protocol for low-latency wireless communication, enabling remote control through both a mobile web interface and a custom Raspberry Pi Pico joystick.
- *Skills Used: Circuit Design, SIMetrix (Circuit Simulation), Arduino IDE Programming, Closed Loop Control Systems*

STM32 Blackjack Game: **August 2025**

- Designed and implemented a functional Blackjack game in C on the STM32F429I Discovery board, utilizing the onboard LCD and touchscreen for real-time game state visualization and user interaction.
- Implemented interrupt-driven touchscreen input handling to process player actions and render card graphics.
- Implemented low-level peripheral control, system timing, and hardware register manipulation to manage game logic and display rendering.
- *Skills Used: C, STM32CubeIDE, GPIO / Interrupt Programming*

SKILLS

- **Programming Languages:** C, C++, Python, SQL, Verilog, MATLAB
- **Hardware:** PCB Design / Assembly, Soldering, Oscilloscopes, Multimeters, Circuit Testing, 3D Printing, CAD
- **Embedded Systems:** ESP32-S3, STM32, Raspberry Pi, Arduino, GPIO, Interrupts
- **Engineering Software:** Altium Designer, ESP-IDF, LVGL, STM32CubeIDE, Arduino IDE, SPICE, Git

ADDITIONAL WORK EXPERIENCE

- **Produce Clerk** | King Soopers (Erie, CO) | *May 2025 – April 2026*
- **Food Court Clerk** | Costco Wholesale (Longmont, CO) | *May 2024 – August 2024*
- **Courtesy Clerk** | Sprouts Farmers Market (Lafayette, CO) | *May 2022 – November 2022*